Basic Web Page Design

Introduction to HTML, CSS, and Bootstrap

source: w3schools.com

Introduction to HTML

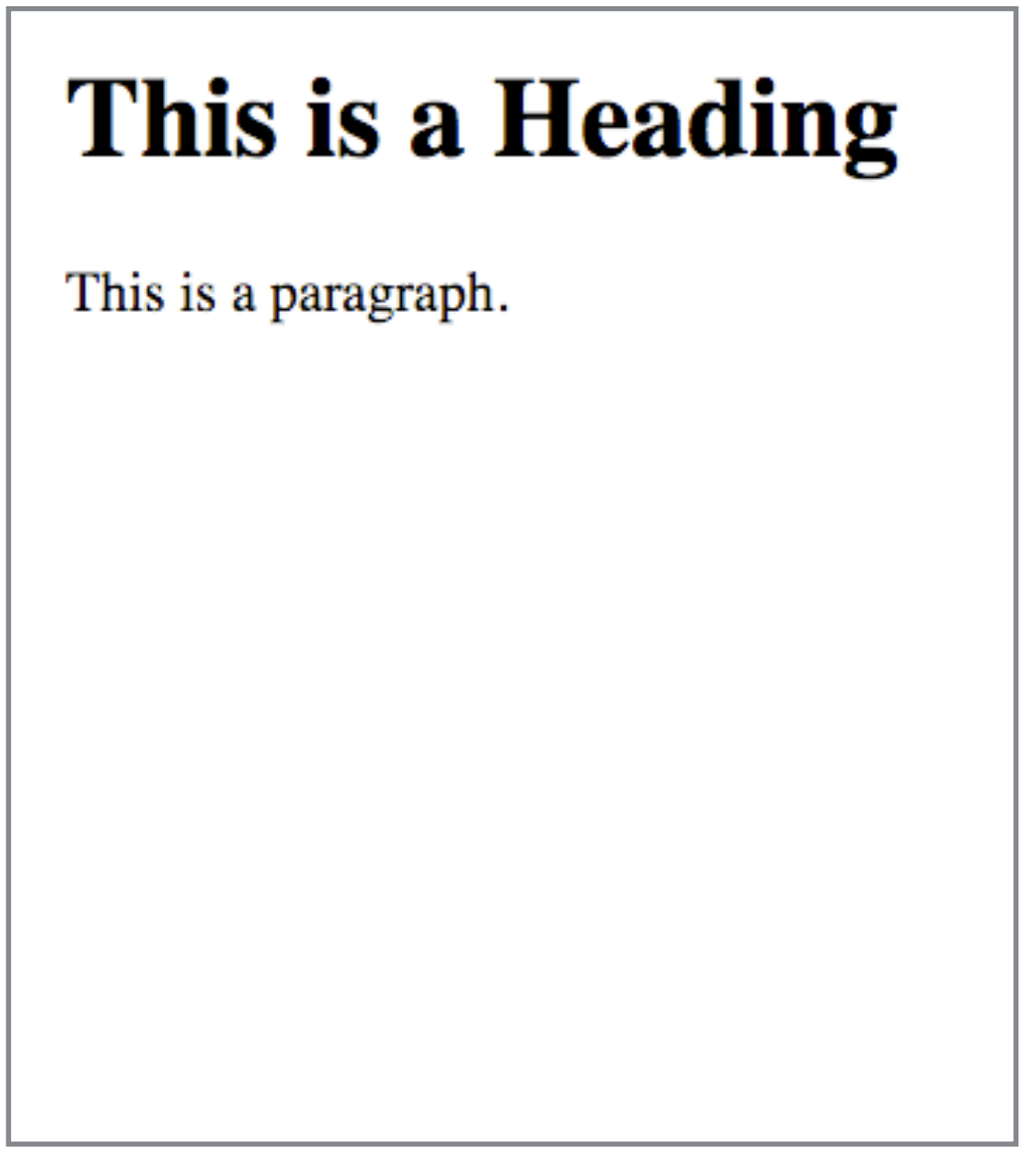


Basic Example

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

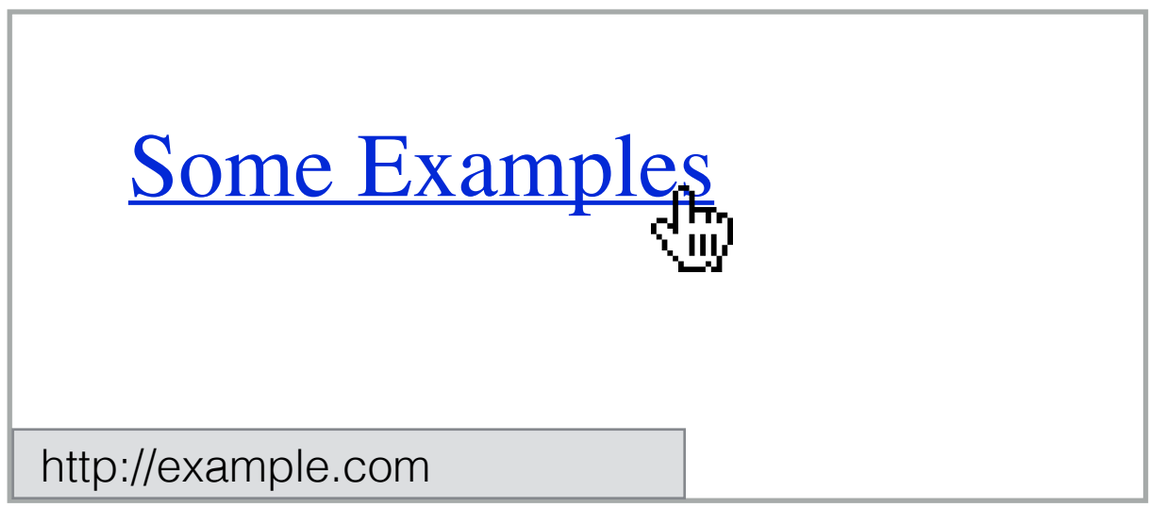
<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```



Elements, Tags, Attributes

```
<a href="http://example.com">Some Examples</a>
```



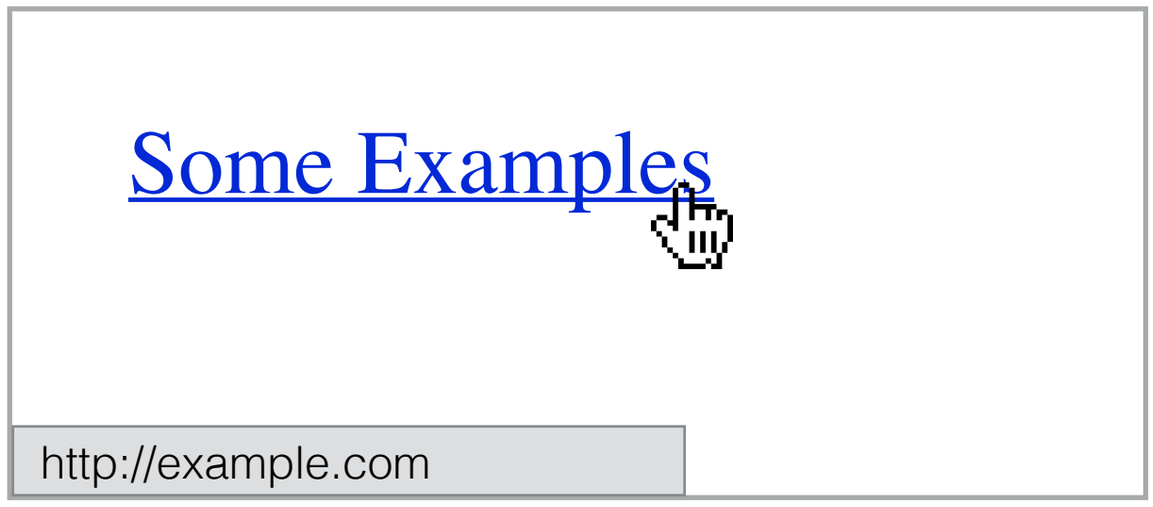
Elements, Tags, Attributes

attribute

```
<a href="http://example.com">Some Examples</a>
```

Tag

Element



Document Structure

```
<html>

  <head>

    <title>Page title</title>

  </head>

  <body>

    <h1>This is a heading</h1>

    <p>This is a paragraph.</p>

    <p>This is another paragraph.</p>

  </body>

</html>
```

head Element

```
<head>

<title>HTML Head Element</title>

<meta charset="UTF-8">
<meta name="description" content="Free Web tutorials">
<meta name="keywords" content="HTML,CSS,JavaScript">
<meta name="author" content="Hege Refsnes">

</head>
```

is shown in the search results pages

used by the search engine ranking algorithms

We will see about <style> and <link> tags later

Coding Conventions

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

Use correct document type

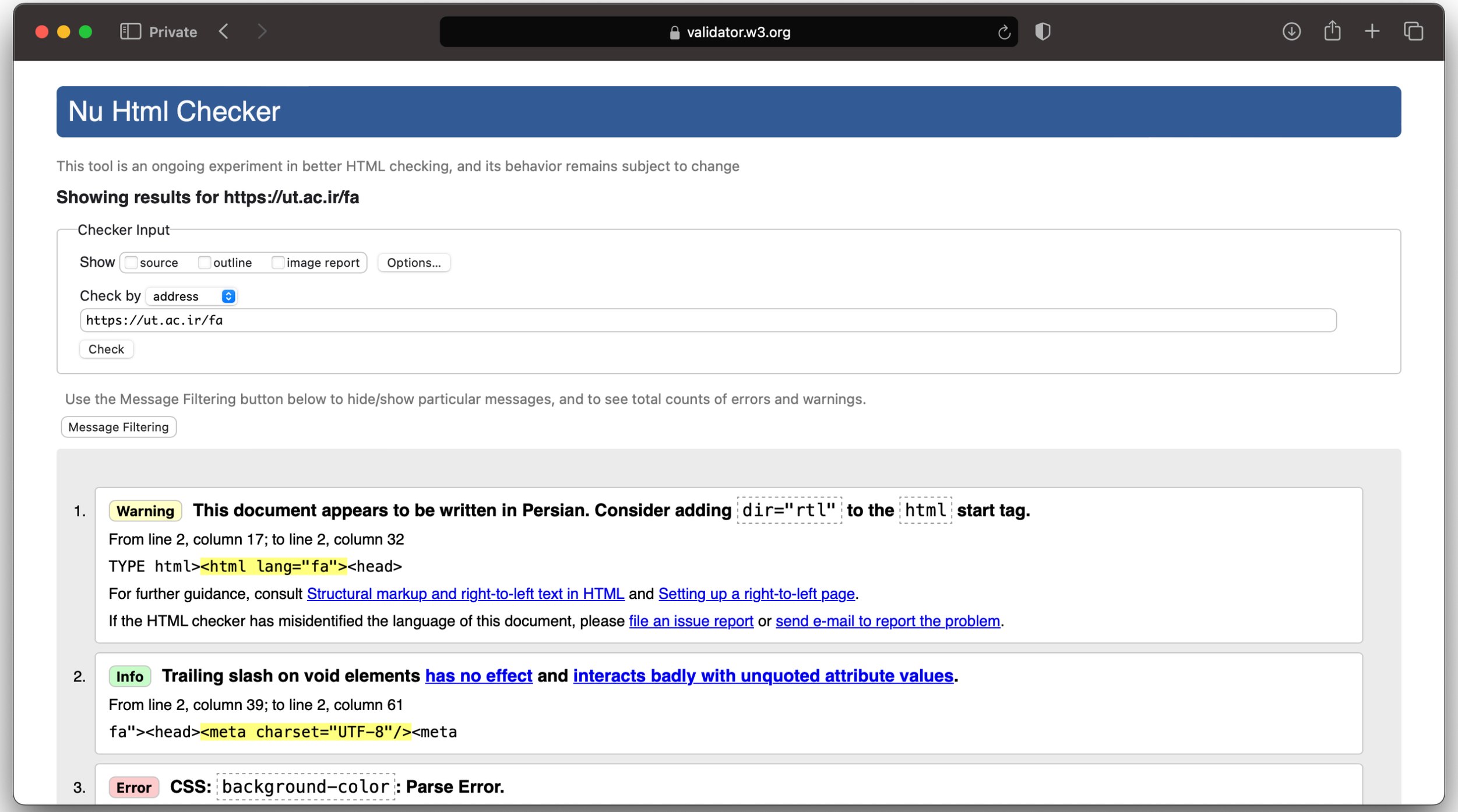
User lower-case tag names

Close all elements

read more at: http://www.w3schools.com/html/html5_syntax.asp

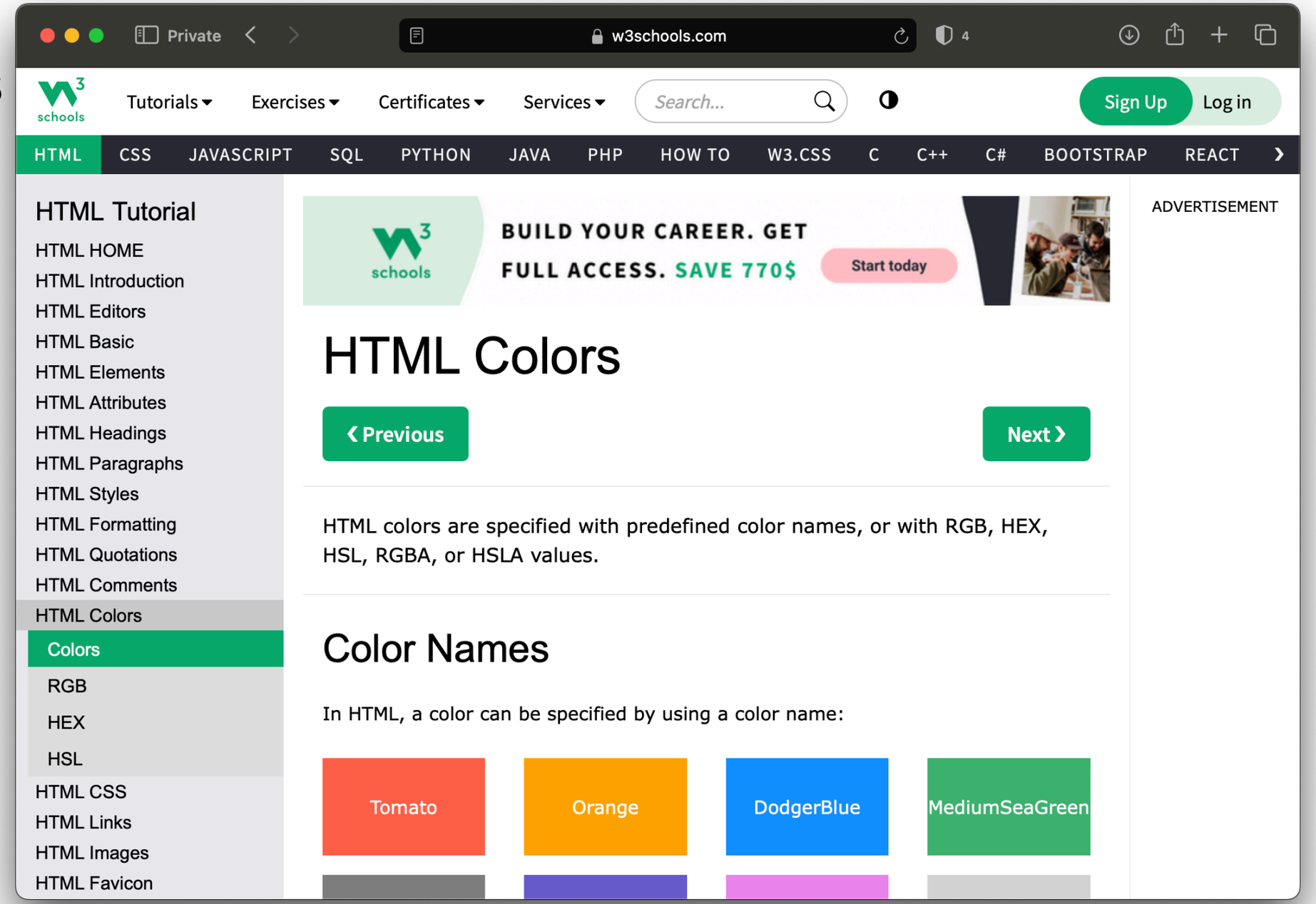
HTML Validators

validator.w3.org



Do It Yourself

- Headings
- Paragraphs
- Styles
- Formatting
- Colors
- Images
- Links
- …



Styles

```
<p style="color:blue; padding:10px; width:250px">
  So, so you think you can tell<br/>
  Heaven from Hell,<br/>
  Blue skies from pain.<br/>
  Can you tell a green field<br/>
  From a cold steel rail?<br/>
  A smile from a veil?<br/>
  Do you think you can tell?
</p>
```

Property	Value
color	blue
padding	10px
width	250px

Tables

```
<table style="width:100%">
  <tr>
    <td>Jill</td>
    <td>Smith</td>
    <td>50</td>
  </tr>
  <tr>
    <td>Eve</td>
    <td>Jackson</td>
    <td>94</td>
  </tr>
  <tr>
    <td>John</td>
    <td>Doe</td>
    <td>80</td>
  </tr>
</table>
```

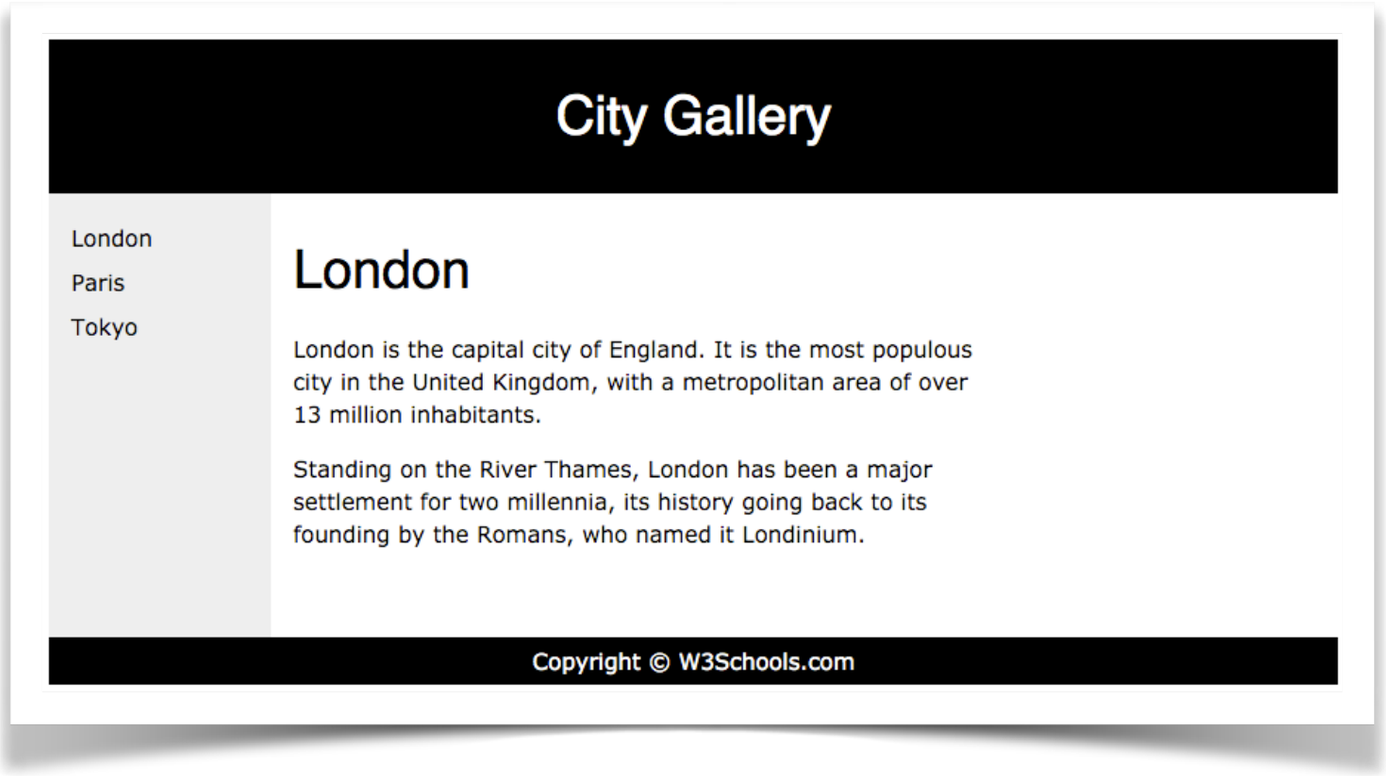
Jill	Smith	50
Eve	Jackson	94
John	Doe	80

Layouts


City Gallery
London
Paris
Tokyo
London
London is the capital city of England. It is the most populous city in the United Kingdom, with a metropolitan area of over 13 million inhabitants.
Standing on the River Thames, London has been a major settlement for two millennia, its history going back to its founding by the Romans, who named it Londinium.
Copyright © W3Schools.com

```
<div id="top">
<h1>City Gallery</h1>
</div>

<div id="menu">
London<br/>
Paris<br/>
Tokyo
</div>
```




```
<div id="main">
<h1>London</h1>
<p>London is the capital city of England. …</p>
</div>

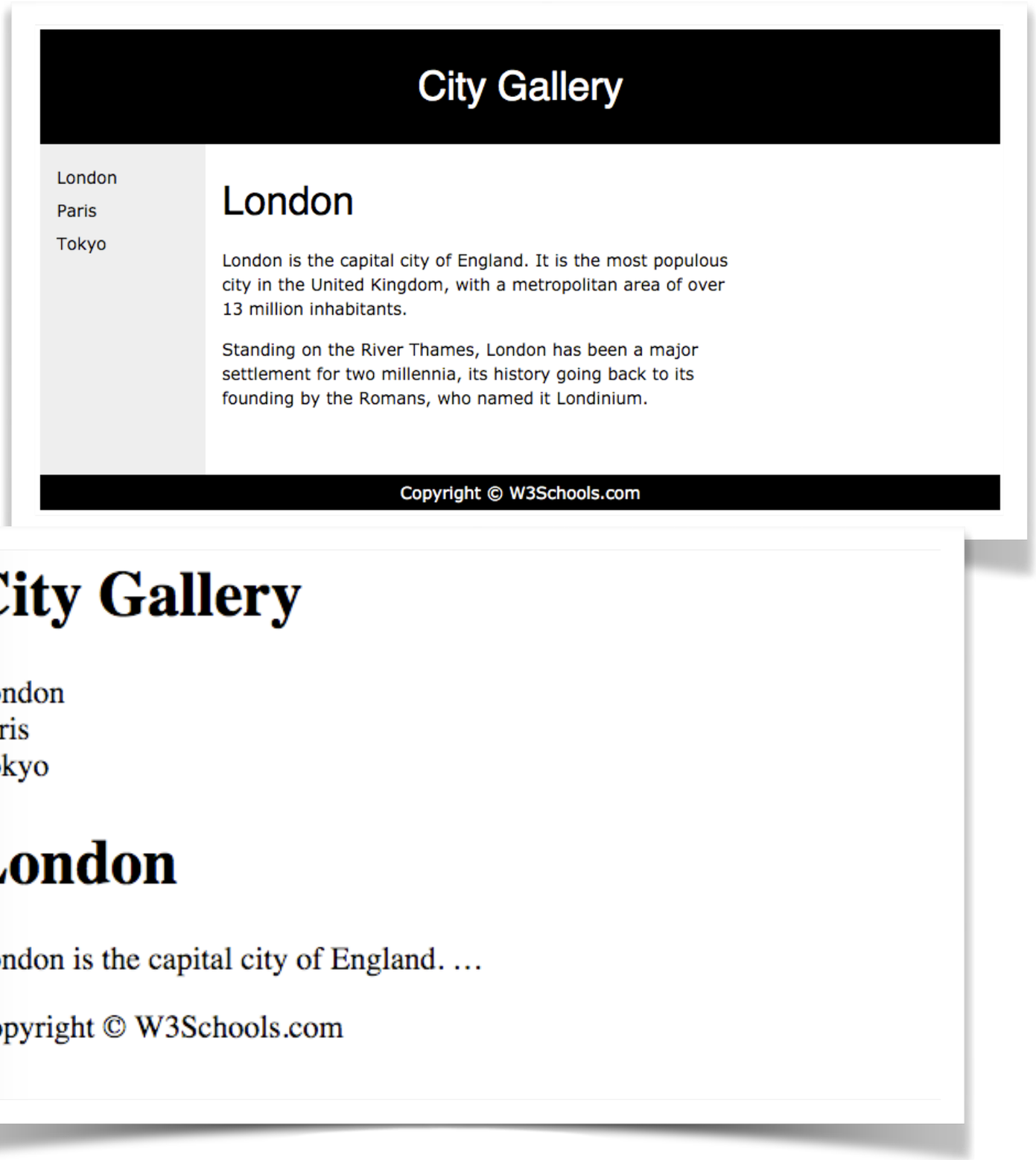
<div id="footer">
Copyright © W3Schools.com
</div>
```

```
<div id="top">
<h1>City Gallery</h1>
</div>

<div id="menu">
London<br/>
Paris<br/>
Tokyo
</div>

<div id="main">
<h1>London</h1>
<p>London is the capi
</div>

<div id="footer">
Copyright © W3Schools
</div>
```

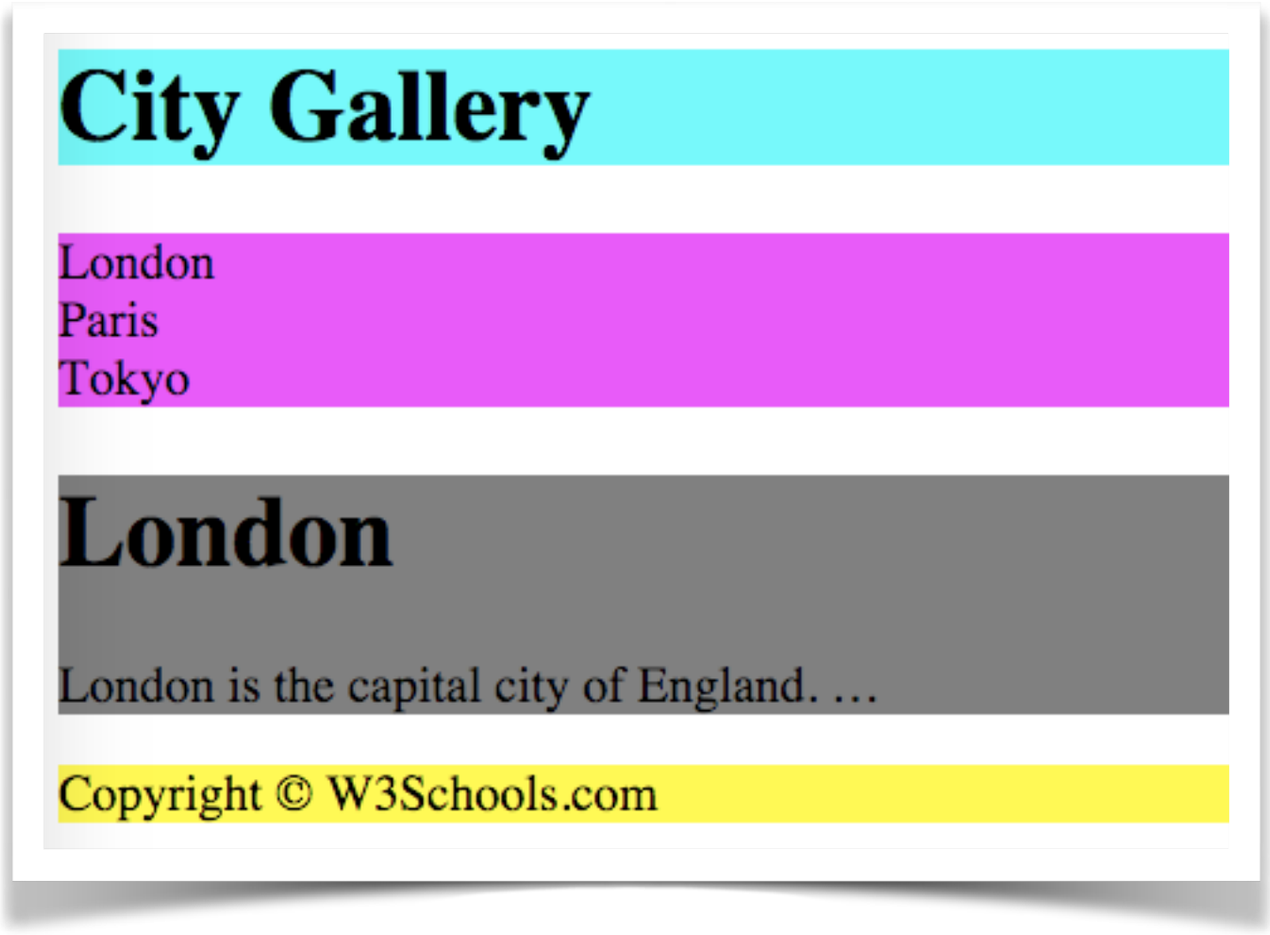


```
<div id="top">
<h1>City Gallery</h1>
</div>

<div id="menu">
London<br/>
Paris<br/>
Tokyo
</div>

<div id="main">
<h1>London</h1>
<p>London is the capital city of England. …</p>
</div>

<div id="footer">
Copyright © W3Schools.com
</div>
```



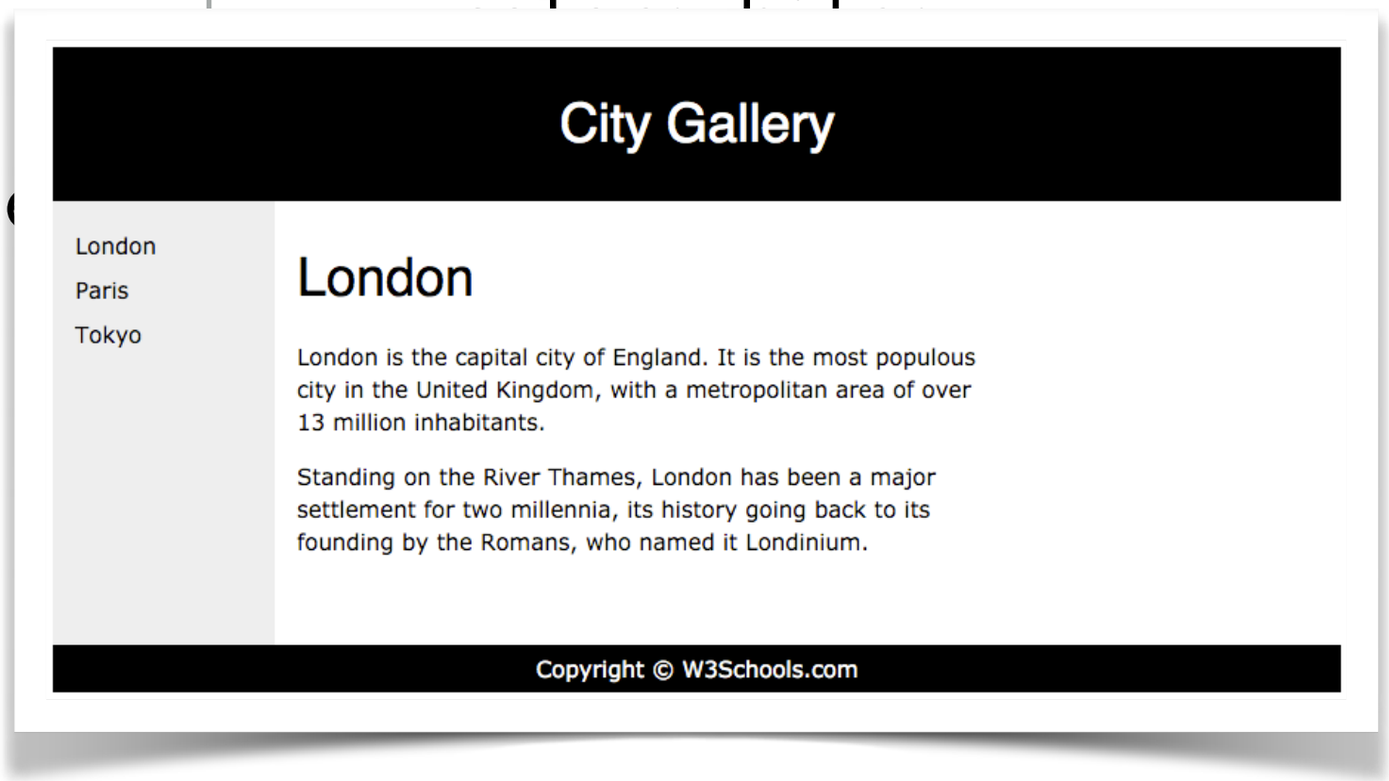
Adding Style

```
<style>
#top {
    background-color:black;
    color:white;
    text-align:center;
    padding:5px;
}
#menu {
    line-height:30px;
    background-color:#eeeeee;
    height:300px;
    width:100px;
    float:left;
    padding:5px;
}
#main {
    width:350px;
    float:left;
    padding:10px;
}
#footer {
    background-color:black;
    color:white;
    clear:both;
    text-align:center;
    padding:5px;
}
</style>
```

Adding Style

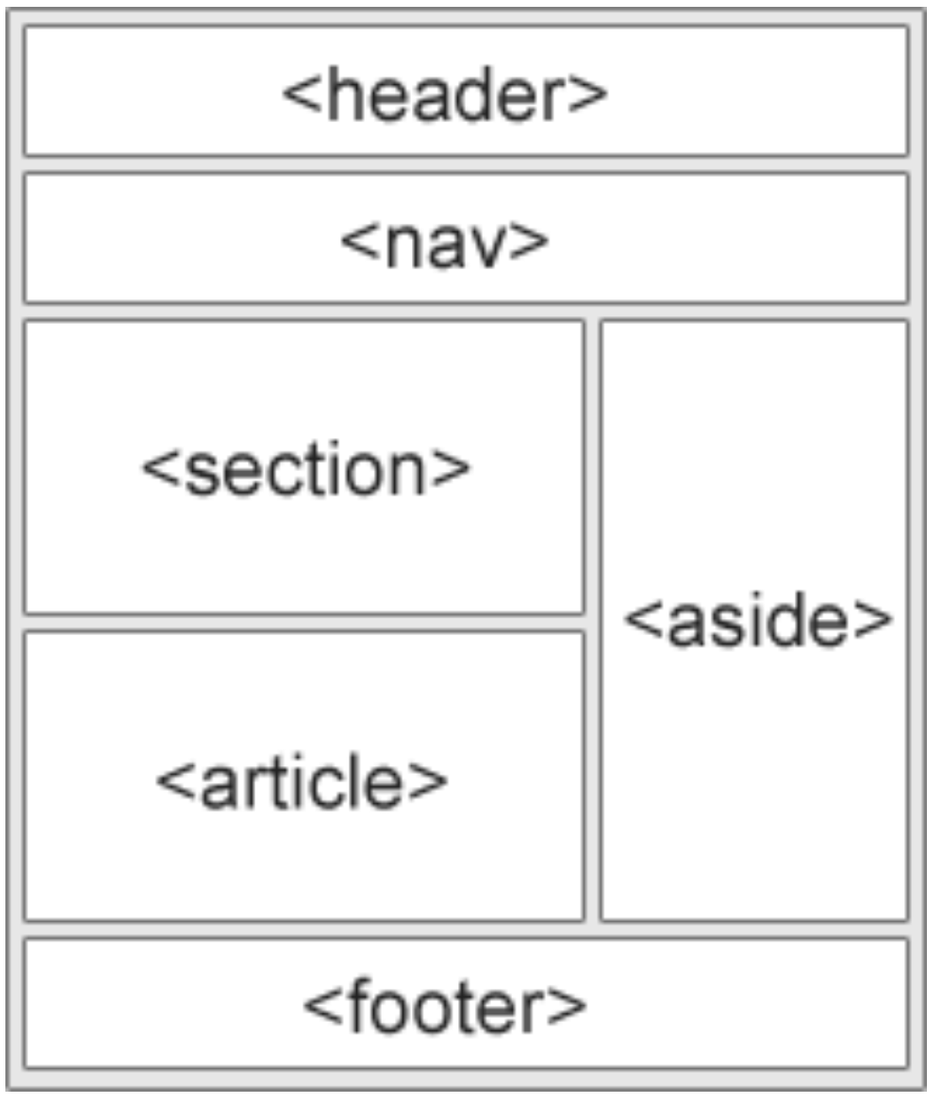
```
<style>
#top {
    background-color:black;
    color:white;
    text-align:center;
    padding:5px;
}
#menu {
    line-height:30px;
    background-color:#eeee
    height:300px;
    width:100px;
    float:left;
    padding:5px;
}
```

```
#main {
    width:350px;
    float:left;
    padding:10px;
}
#footer {
    background-color:black;
```



HTML5 Semantic Elements

- <article>
- <aside>
- <details>
- <figcaption>
- <figure>
- <footer>
- <header>
- <main>
- <mark>
- <nav>
- <section>
- <summary>
- <time>



HTML Forms

```
<form action="action_page.php">
  <fieldset>
    <legend>Personal information:</legend>
    First name:<br>
    <input type="text" name="firstname" value="Mickey"><br>
    Last name:<br>
    <input type="text" name="lastname" value="Mouse"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>
```

Personal information:

First name:

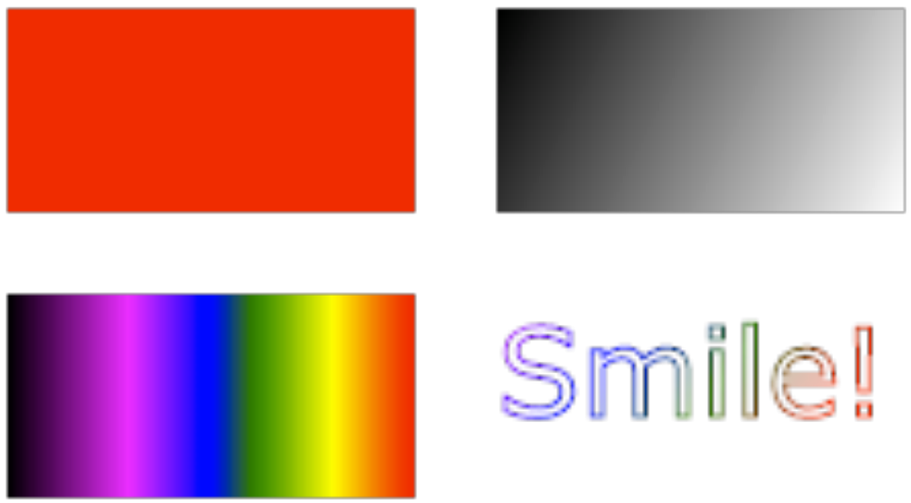
Mickey

Last name:

Mouse

Submit

HTML5 Graphics



<canvas> element

Scalable Vector Graphics (SVG)



```
<svg width="300" height="200">
  <polygon points="100,10 40,198 190,78 10,78 160,198"
  style="fill:lime;stroke:purple;stroke-width:5;fill-rule:evenodd;" />
</svg>
```

HTML5 Multimedia

```
<video width="320" height="240" autoplay>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.
</video>
```

Supports common audio/video formats



HTML5 APIs

- Geolocation
  Determining users's location
- Drag/Drop
  Drag and drop on the web pages
- Local Storage
  Store data locally within the users's browser (better than cookies)
- App Cache
  Make your application available offline
- Web Workers
  JavaScript job running in the background
- Server-Sent Events
  Allowing a web page to receive updates from a server

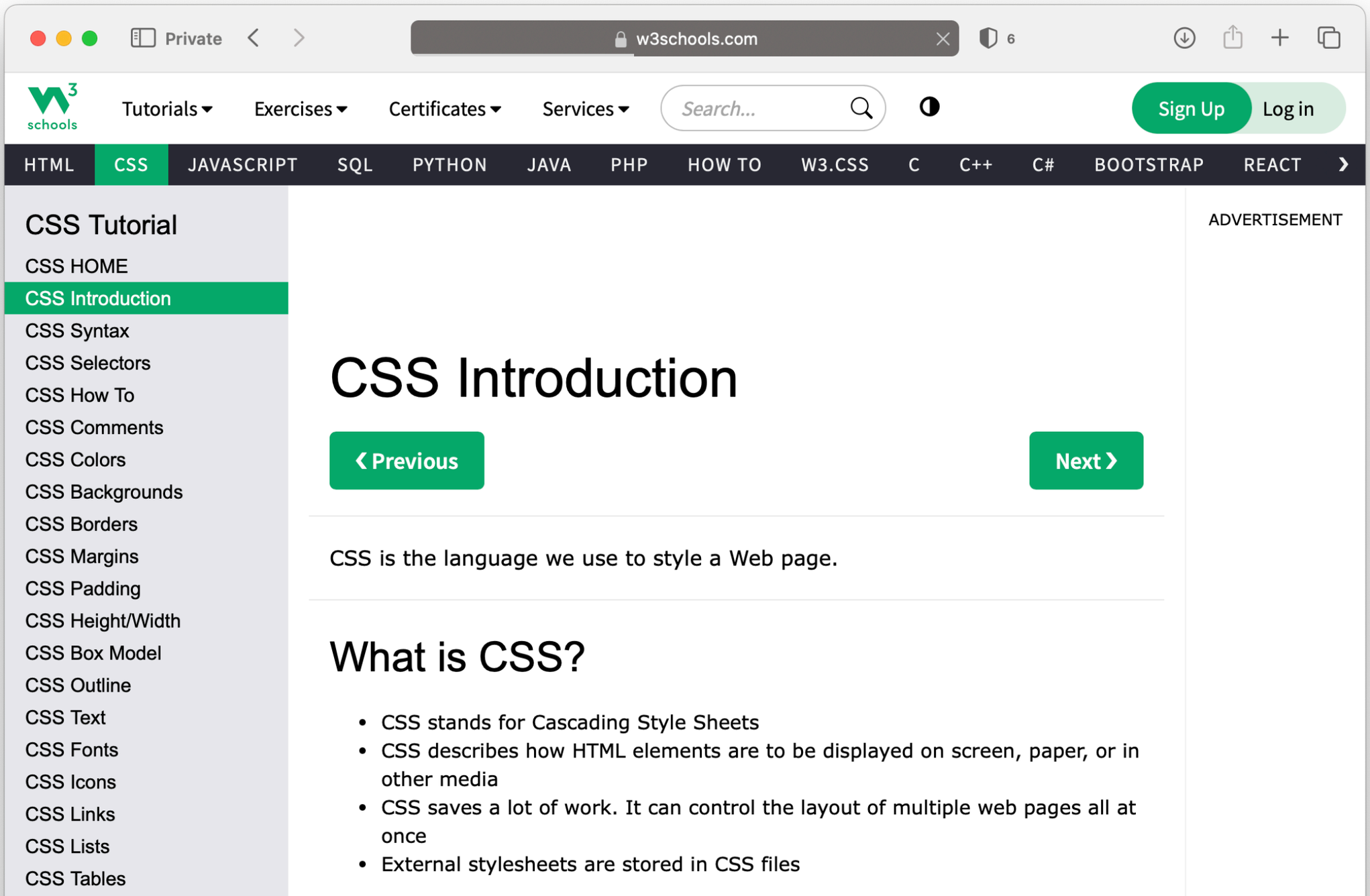
Read on w3schools.com

Introduction to CSS



Same Content, Different Looks

http://www.w3schools.com/css/css_intro.asp



Basic CSS Example

```
body {
    background-color: #d0e4fe;
}

h1 {
    color: orange;
    text-align: center;
}

p {
    font-family: "Times New Roman";
    font-size: 20px;
}
```

Basic CSS Example

```
body {
    background-color: #d0e4fe;
}

h1 {
    color: orange;
    text-align: center;
}

p {
    font-family: "Times New Roman";
    font-size: 20px;
}
```

Selector

Rule

Declaration

Property

Value

Basic Selectors

Element Selector	`p { … }`	`<p>`
Id Selector	`#par1 { … }`	`<p id=“par1”>`
Class Selector	`.note { … }`	`<p class=“note”>` `<td class=“note”>`
Class Selector (limited to an element)	`p.note { … }`	`<p class=“note”>` ~~`<td class=“note”>`~~
Universal Selector	`* { … }`	Every element!

Descendant Selectors

```
.note h2 { … }
```

```
#main h2 { … }
```

```
<div class="note">
    <h2> … </h2>
</div>
```

```
<div id="main">
    <h2> … </h2>
</div>
```

Pseudo-Classes

```
a:link { color: #FF0000; }
a:visited { color: #00FF00; }
a:hover { color: #FF00FF; }
a:active { color: #0000FF; }
```

```
p i:first-child {
    color: blue;
}
```

Pseudo-Classes

```
<!DOCTYPE html>
<html>
<head>
<style>
p i:first-child {
  background: yellow;
}
</style>
</head>
<body>

<p>I am a <i>strong</i> man. I am a
<i>strong</i> man.</p>
<p>I am a <i>strong</i> man. I am a
<i>strong</i> man.</p>
```

I am a *strong* man. I am a *strong* man.

I am a *strong* man. I am a *strong* man.

```
<!DOCTYPE html>
<html>
<head>
<style>
p:first-child i {
  background: yellow;
}
</style>
</head>
<body>

<p>I am a <i>strong</i> man. I am a
<i>strong</i> man.</p>
<p>I am a <i>strong</i> man. I am a
<i>strong</i> man.</p>
```

I am a *strong* man. I am a *strong* man.

I am a *strong* man. I am a *strong* man.

Inserting CSS - External

some.html

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

mystyle.css

```
body {
    background-color: lightblue;
}

h1 {
    color: navy;
    margin-left: 20px;
}
```

Inserting CSS - <style> Tag

some.html

```
<head>
   <style>
      body {
          background-color: linen;
      }

      h1 {
          color: maroon;
          margin-left: 40px;
      }
   </style>
</head>
```

Inserting CSS - Inline Styles

some.html

```
<h1 style="color:blue;margin-left:30px;">
  This is a heading.
</h1>
```

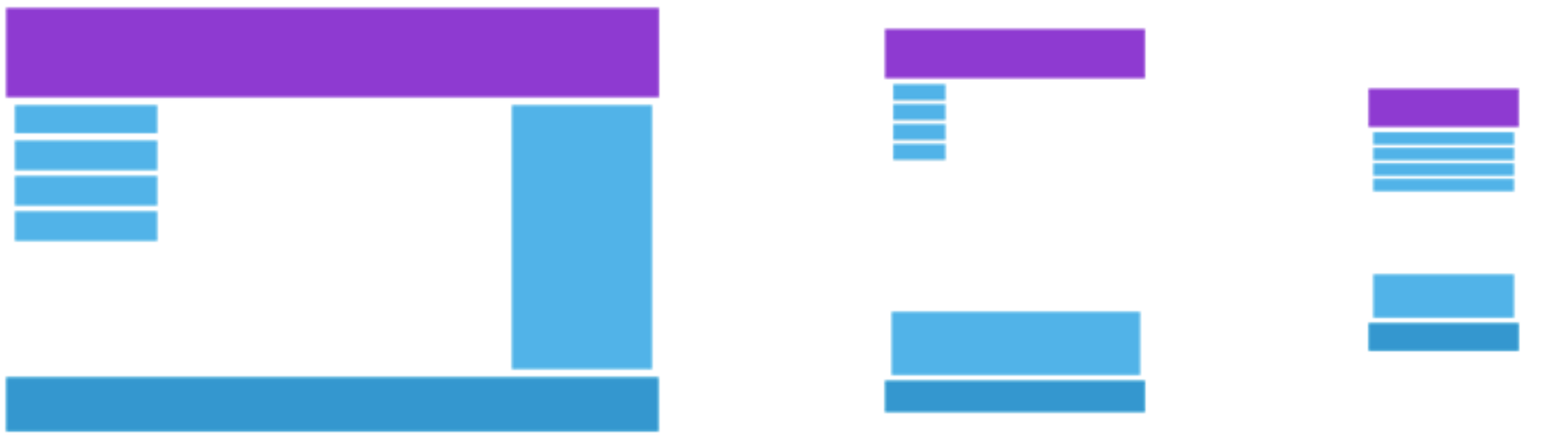
Cascading Styles

The order of applying styles (increasing priority):

1. Browser Defaults
2. External Style Sheets (Linked or Imported)
3. Internal Style Sheets (Embedded)
4. Inline Styles (inside the HTML element)

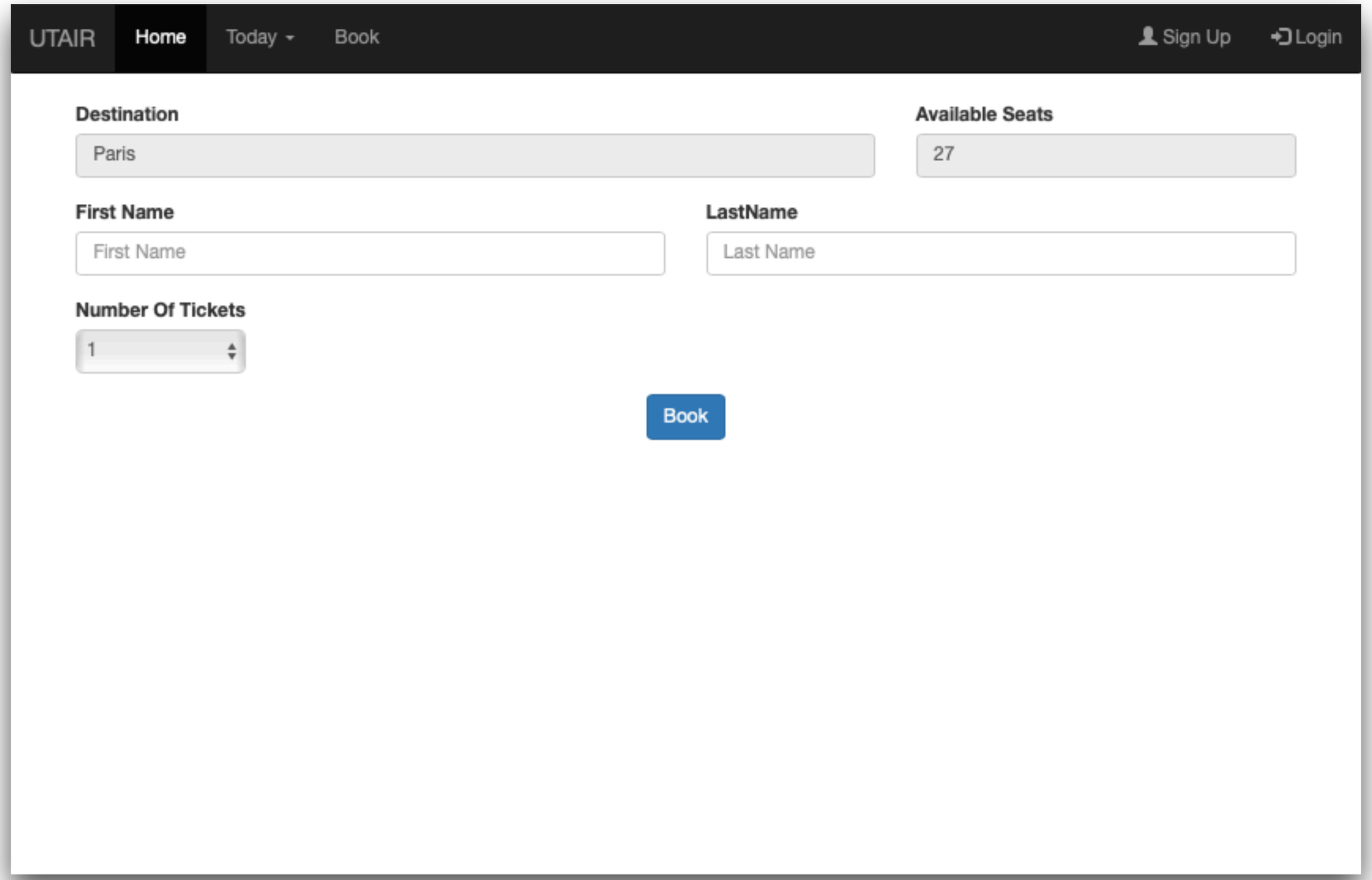
Responsive Web Design

Make your web page look good on all devices

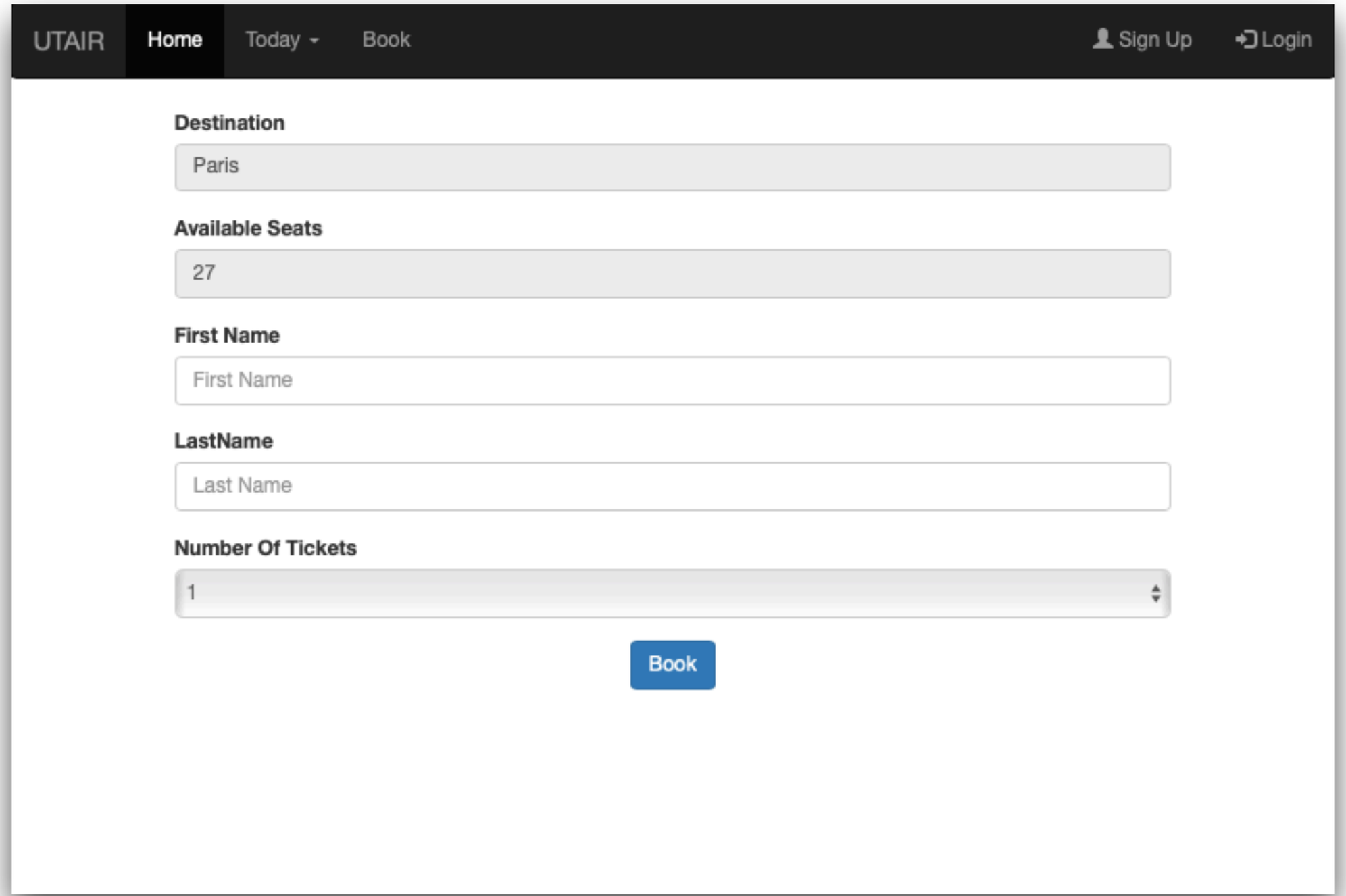


Desktop Tablet Phone

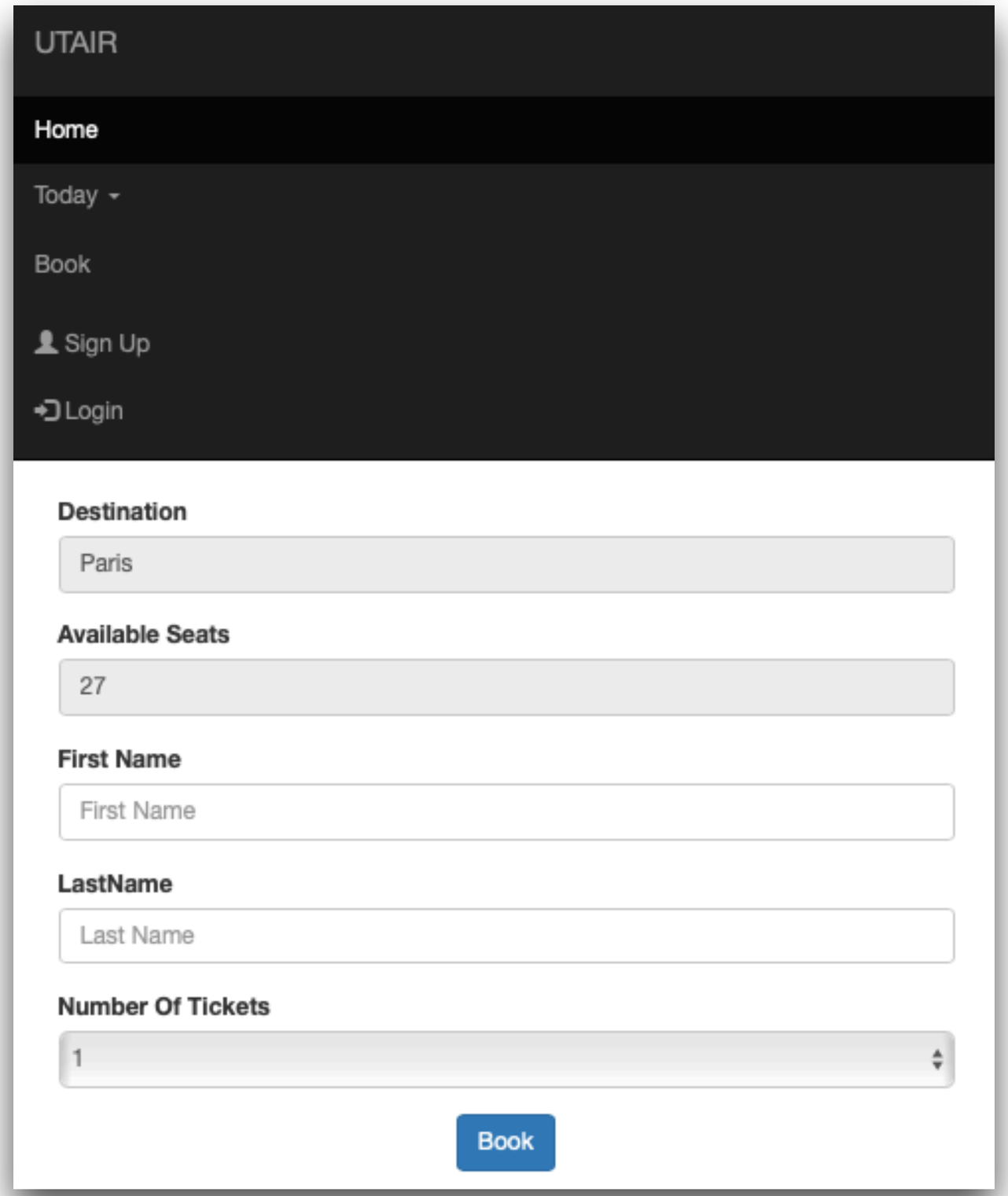
UTAir in Wide Screens



UTAir in Tablet Sized Screens



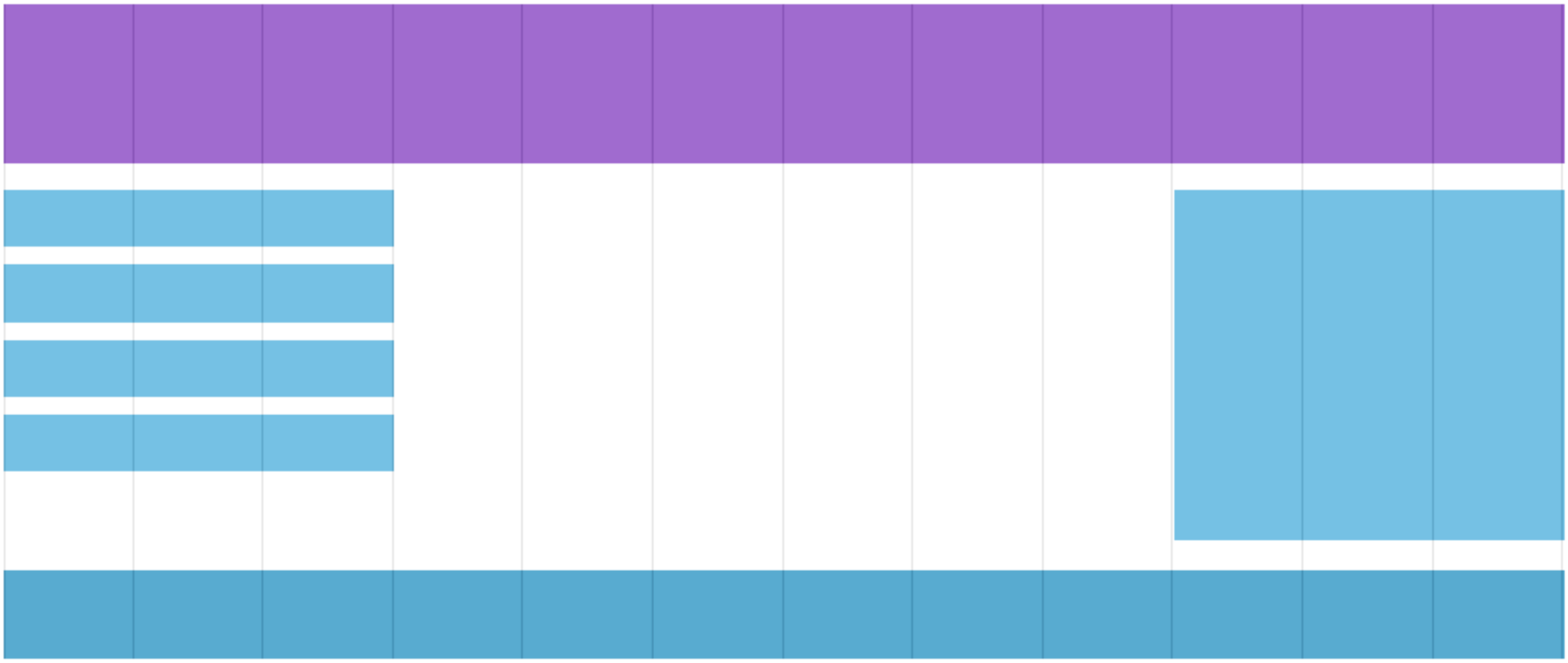
UTAir in Cellphones Screen



RWD - General Rules

1. Do not use large fixed width elements
2. Do not rely on a particular viewport width
3. Use CSS media queries

RWD Grid



http://www.w3schools.com/css/css_rwd_grid.asp shows how to build an RWD grid.

CSS Media Queries

If the browser window is smaller than 500px,
the background color will change to light blue:

```
@media only screen and (max-width: 500px) {
    body {
        background-color: lightblue;
    }
}
```

Defining Breakpoints

```
/* For desktop: */
.col-1 {width: 8.33%;}
.col-2 {width: 16.66%;}
.col-3 {width: 25%;}
.col-4 {width: 33.33%;}
.col-5 {width: 41.66%;}
.col-6 {width: 50%;}
/* other column definitions */

@media only screen and (max-width: 768px) {
    /* For mobile phones: */
    [class*="col-"] {
        width: 100%;
    }
}
```





Mobile-First Design

```
/* For mobile phones: */
[class*="col-"] {
    width: 100%;
}

@media only screen and (min-width: 768px) {
    /* For desktop: */
    .col-1 {width: 8.33%;}
    .col-2 {width: 16.66%;}
    .col-3 {width: 25%;}
    .col-4 {width: 33.33%;}
    .col-5 {width: 41.66%;}
    .col-6 {width: 50%;}
    /* other column definitions */
}
```

More Breakpoints

```
/* For mobile phones: */
[class*="col-"] {
    width: 100%;
}
@media only screen and (min-width: 600px) {
    /* For tablets: */
    .col-m-1 {width: 8.33%;}
    .col-m-2 {width: 16.66%;}
    /* other column definitions */
}
@media only screen and (min-width: 768px) {
    /* For desktop: */
    .col-1 {width: 8.33%;}
    .col-2 {width: 16.66%;}
    /* other column definitions */
}
```

HTML Example for RWD

```
<div class="row">
<div class="col-3 col-m-3">...</div>
<div class="col-6 col-m-9">...</div>
<div class="col-3 col-m-12">...</div>
</div>
```

Introduction to Bootstrap



Bootstrap Framework

- A free front-end framework for web development
- HTML and CSS design templates
- Making mobile-first RWD of web pages easy

Basic Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
        href="http://maxcdn.bootstrapcdn.com/.../bootstrap.min.css">
  <script src="https://ajax.googleapis.com/.../jquery.min.js"/>
  <script src="http://maxcdn.bootstrapcdn.com/.../bootstrap.min.js"/>
</head>
<body>

<div class="container">
  <h1>My First Bootstrap Page</h1>
  <p>This is some text.</p>
</div>

</body>
</html>
```

Basic Bootstrap Grid

span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1
span 4				span 4				span 4			
span 4				span 8							
span 6						span 6					
span 12											

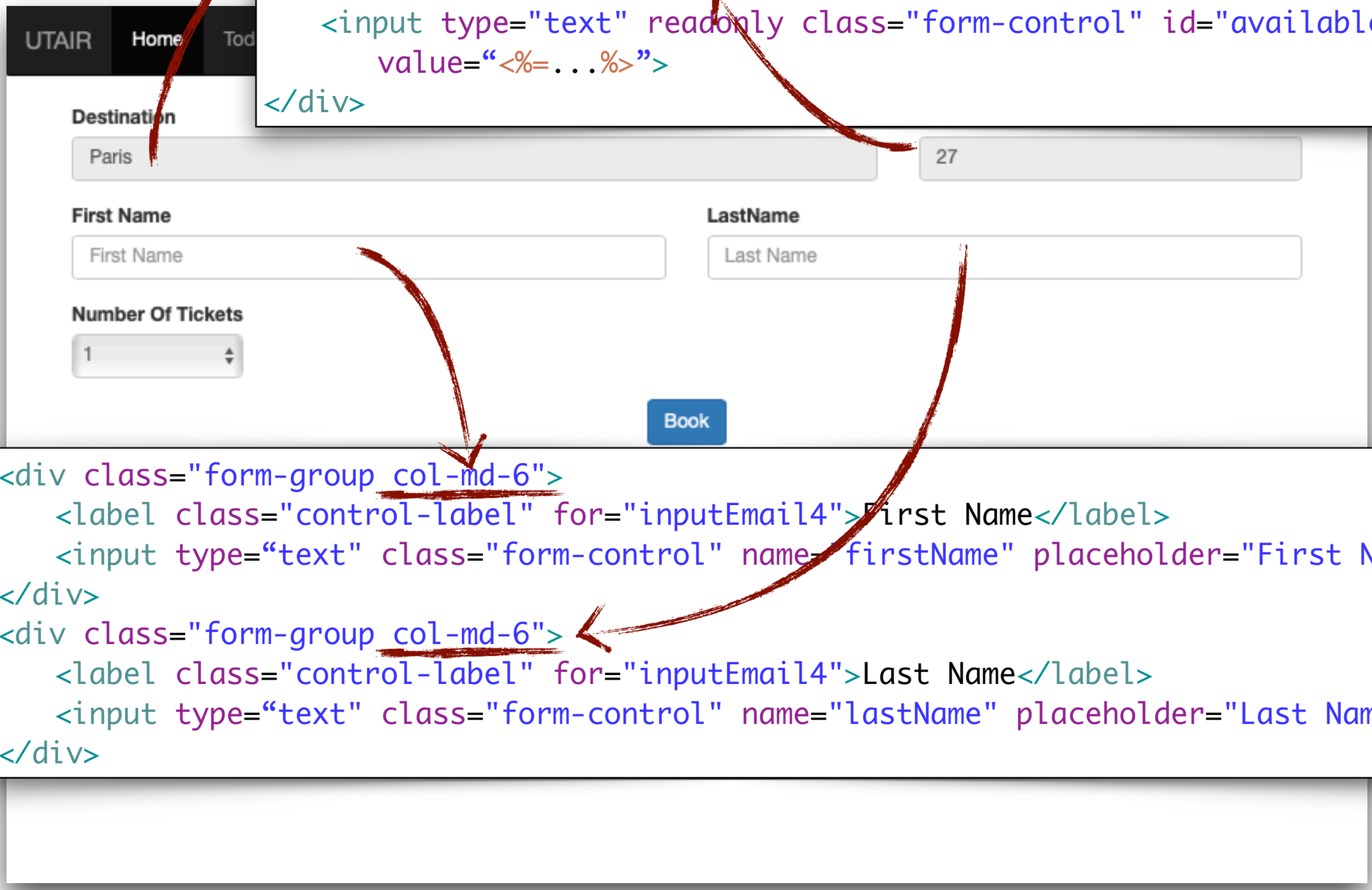
Basic Structure

```
<div class="container">
  <div class="row">
    <div class="col-*-*"></div>
  </div>
  <div class="row">
    <div class="col-*-*"></div>
    <div class="col-*-*"></div>
  </div>
  <div class="row">
    ...
  </div>
</div>
```

```
<div class="row">
  <div class="col-sm-4">.col-sm-4</div>
  <div class="col-sm-8">.col-sm-8</div>
</div>
```

```
<div class="form-group col-md-8">
    <label for="exampleFormControlInput1">Destination</label>
    <input type="text" readonly class="form-control" name="destination"
        value="<%=destination%>">
</div>
```

```
<div class="form-group col-md-4">
    <label for="exampleFormControlInput1">Available Seats</label>
    <input type="text" readonly class="form-control" id="availableSeats"
        value="<%=...%>">
</div>
```




```
<div class="form-group col-md-6">
    <label class="control-label" for="inputEmail4">First Name</label>
    <input type="text" class="form-control" name="firstName" placeholder="First Name">
</div>
<div class="form-group col-md-6">
    <label class="control-label" for="inputEmail4">Last Name</label>
    <input type="text" class="form-control" name="lastName" placeholder="Last Name">
</div>
```

Grid Options

	Extra small devices Phones (<768px)	Small devices Tablets (>=768px)	Medium devices Desktops (>=992px)	Large devices Desktops (>=1200px)
Grid behaviour	Horizontal at all times	Collapsed to start, horizontal above breakpoints	Collapsed to start, horizontal above breakpoints	Collapsed to start, horizontal above breakpoints
Container width	None (auto)	750px	970px	1170px
Class prefix	.col-xs-	.col-sm-	.col-md-	.col-lg-
Number of columns	12	12	12	12
Column width	Auto	~62px	~81px	~97px
Gutter width	30px (15px on each side of a column)	30px (15px on each side of a column)	30px (15px on each side of a column)	30px (15px on each side of a column)
Nestable	Yes	Yes	Yes	Yes
Offsets	Yes	Yes	Yes	Yes
Column ordering	Yes	Yes	Yes	Yes

Applying Styles



learn more on w3schools.com

Bootstrap Templates at startbootstrap.com

